



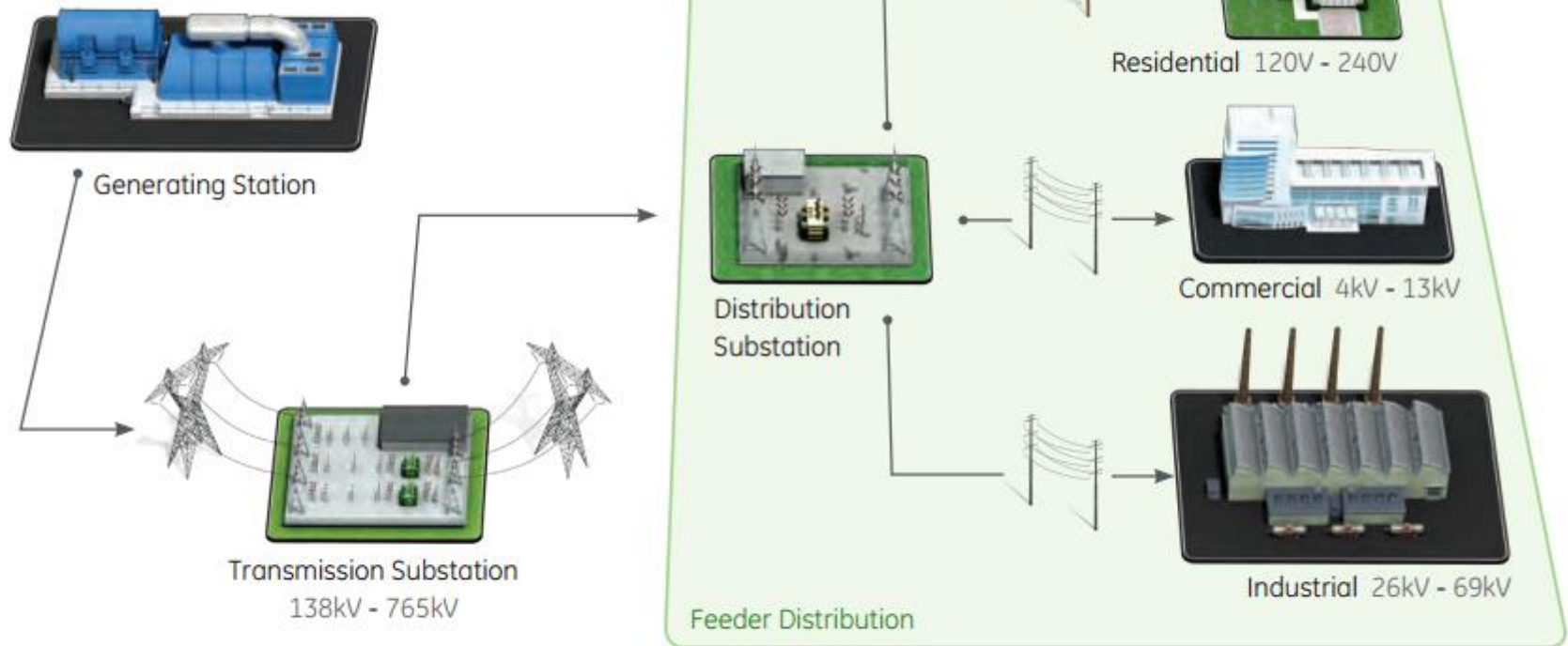
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Distribution of Electrical Energy

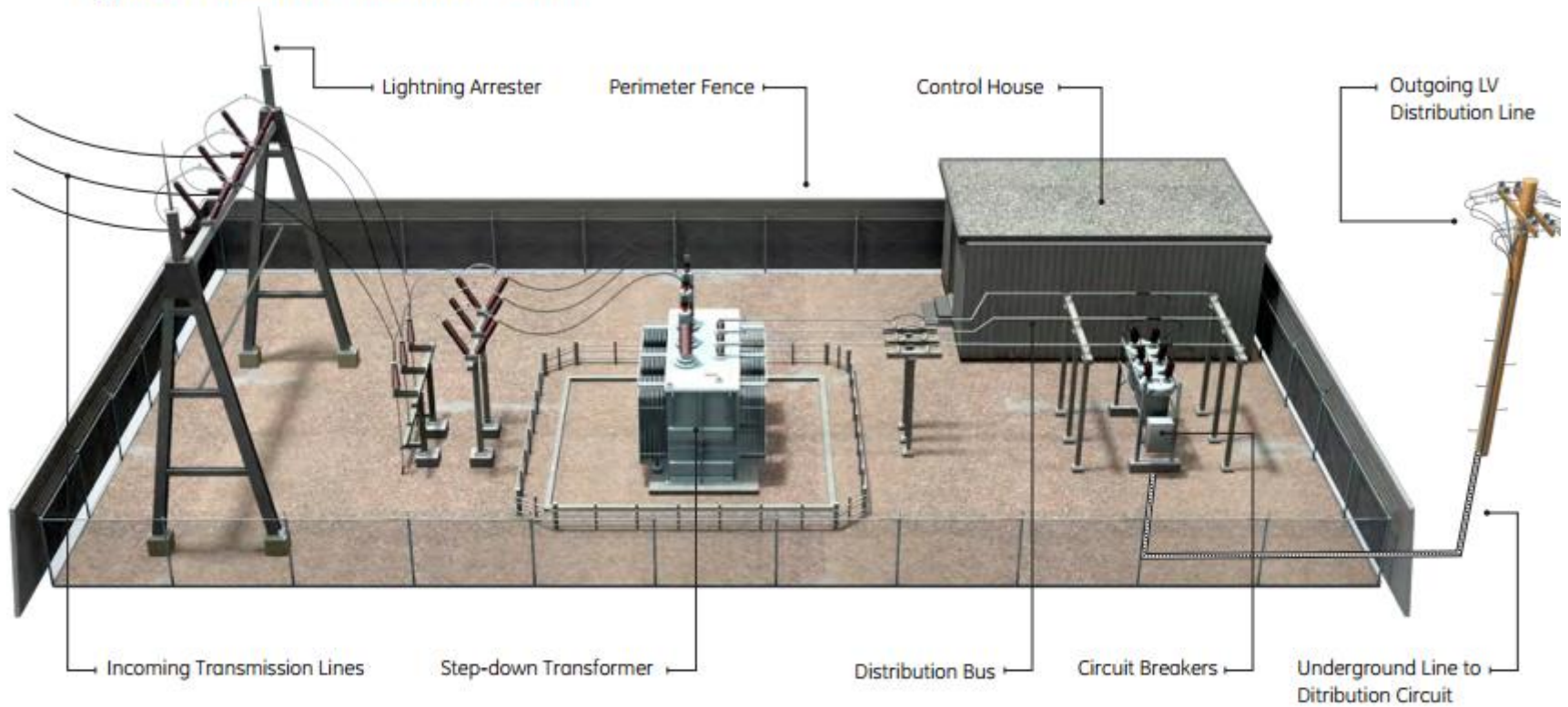
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Distribution in the Power System



Typical Distribution Substation



Primary Distribution

- Distribution circuits come in many different configurations and circuit lengths.
- A feeder is one of the circuits out of the substation.
- The main feeder is the three-phase backbone of the circuit, which is often called the mains or mainline.
- The mainline is normally a modestly large conductor

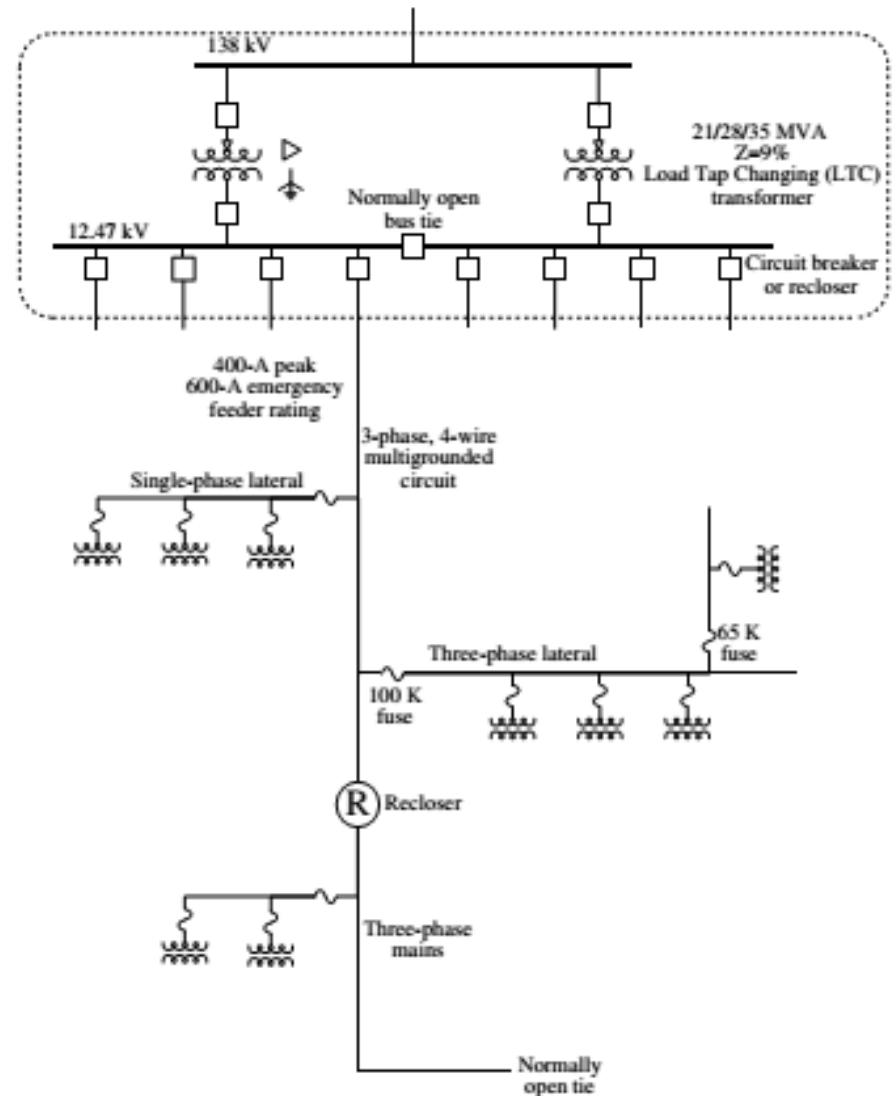


FIGURE 1.3

Typical distribution substation with one of several feeders shown (many lateral taps are left off). (Copyright © 2000, Electric Power Research Institute. 1000419. *Engineering Guide for Integration of Distributed Generation and Storage Into Power Distribution Systems*. Reprinted with permission.)

Primary Distribution

- Branching from the mains are one or more laterals, which are also called taps, lateral taps, branches, or branch lines.
- These **laterals** may be single-phase, two phase, or **three-phase**.
- The laterals normally have fuses to separate them from the mainline if they are faulted.

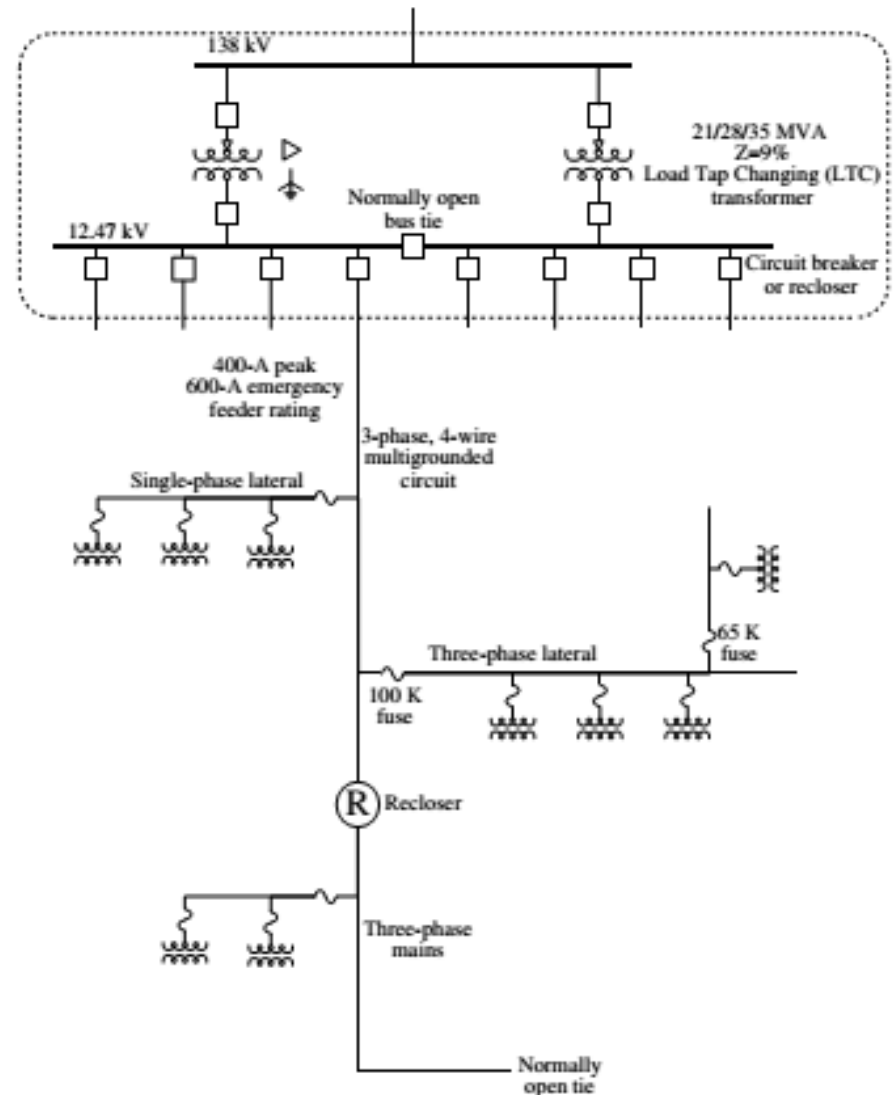


FIGURE 1.3

Typical distribution substation with one of several feeders shown (many lateral taps are left off). (Copyright © 2000, Electric Power Research Institute. 1000419. *Engineering Guide for Integration of Distributed Generation and Storage Into Power Distribution Systems*. Reprinted with permission.)

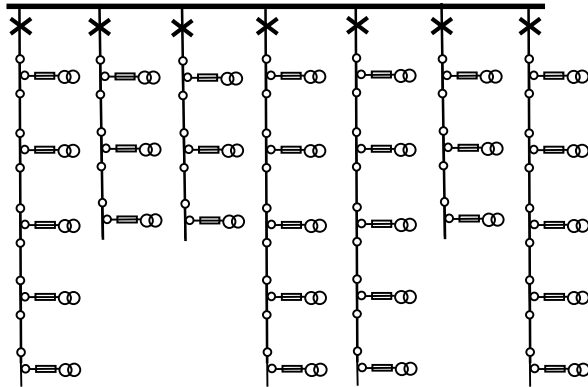
Primary Distribution

- There are several configurations of primary distribution systems.
 - Radial system
 - Loop system
 - Open loop system
 - Closed loop system
 - Network system

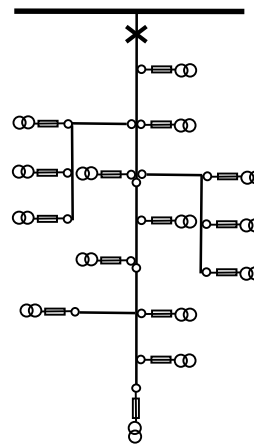
Primary Distribution

radial system

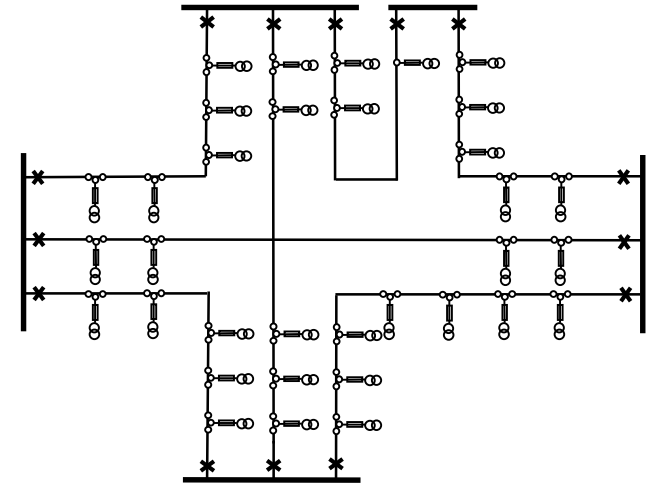
Urban



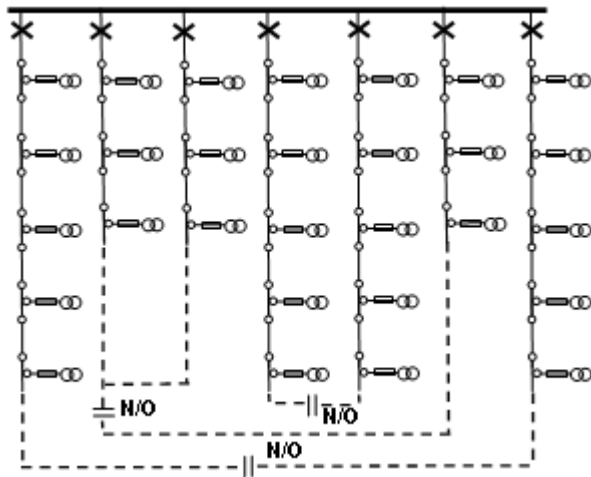
Rural



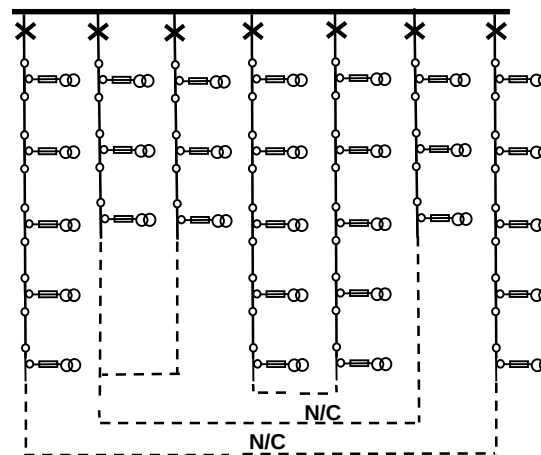
Primary Network system



Open loop system



Closed loop system



Medium Voltage
Distribution
Network
Structures

Primary Distribution

Radial primary distribution system

- The simplest and lowest cost and therefore most common form of primary feeder

- The current magnitude is the greatest in the conductors that leave the substation
- The current magnitude continuously lessens out toward the end of the feeder.

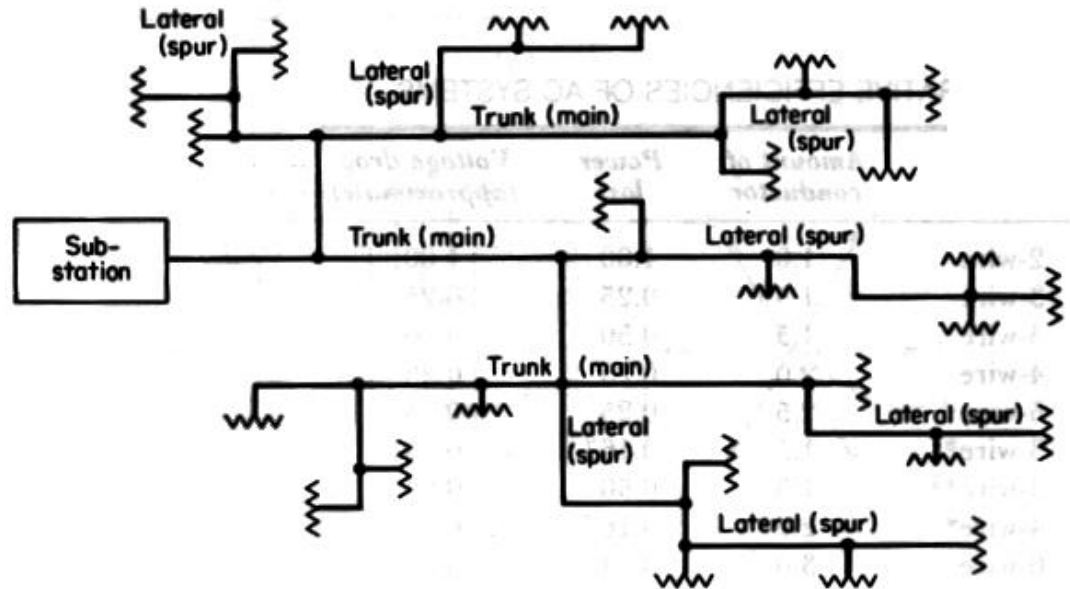
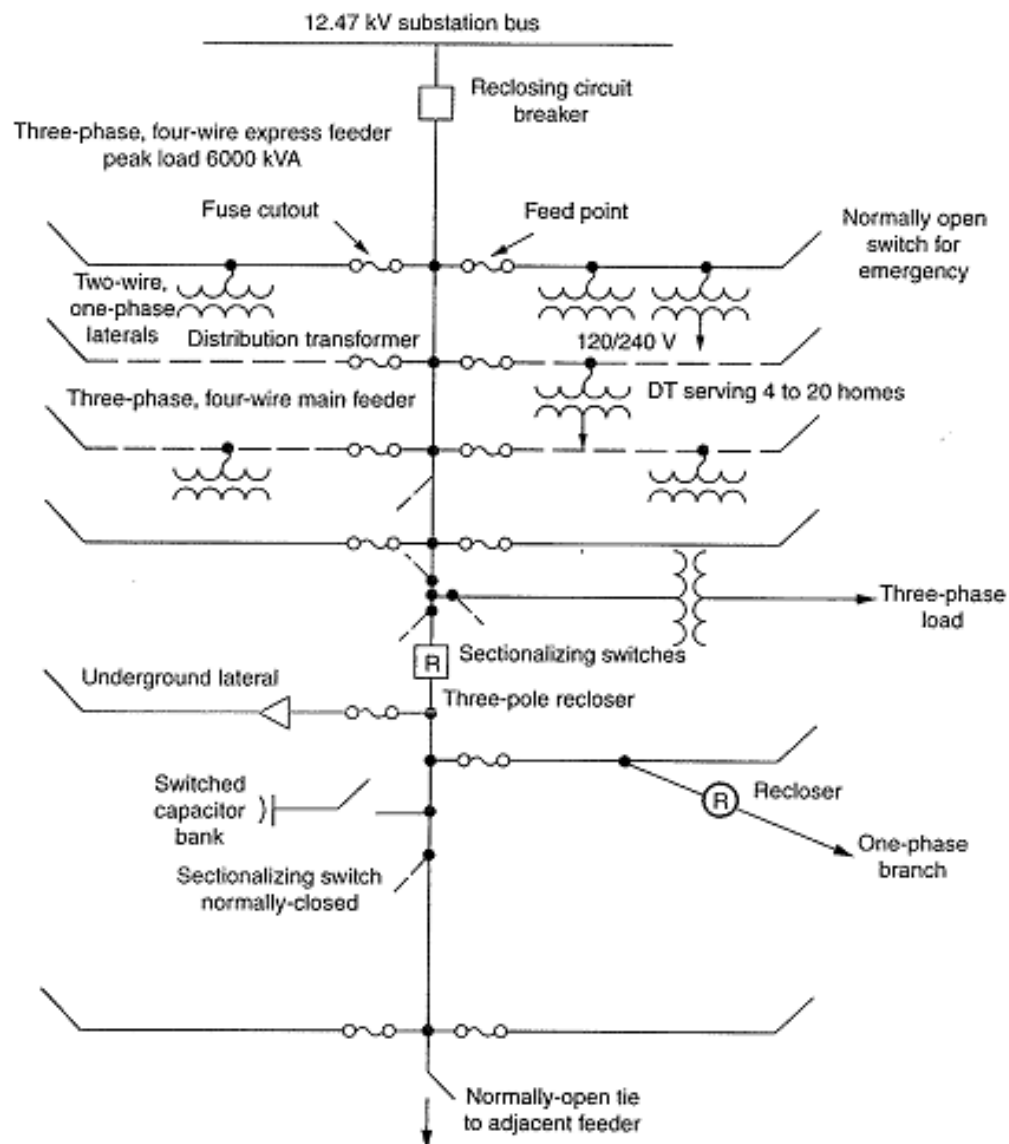


FIG. 2-11 Primary feeder schematic diagram showing trunk or main feeds and laterals or spurs.



Residential area: Approximately 1000 homes per square mile
 Feeder area: 1 to 4 mi² depending on load density
 15 to 30 single-phase laterals per feeder
 150 to 500 MVA short-circuit available at substation bus

FIGURE 5.1 One-line diagram of typical primary distribution feeders. (From Fink, D. G., and H. W. Beaty, *Standard Handbook for Electrical Engineers*, 11th ed., McGraw-Hill, New York, 1978. With permission.)

The reliability of service continuity of the radial primary feeders is low. A fault occurrence at any location on the radial primary feeder causes a power outage for every consumer on the feeder unless the fault can be isolated from the source by a disconnecting device such as a fuse, sectionalizer, disconnect switch, or recloser.

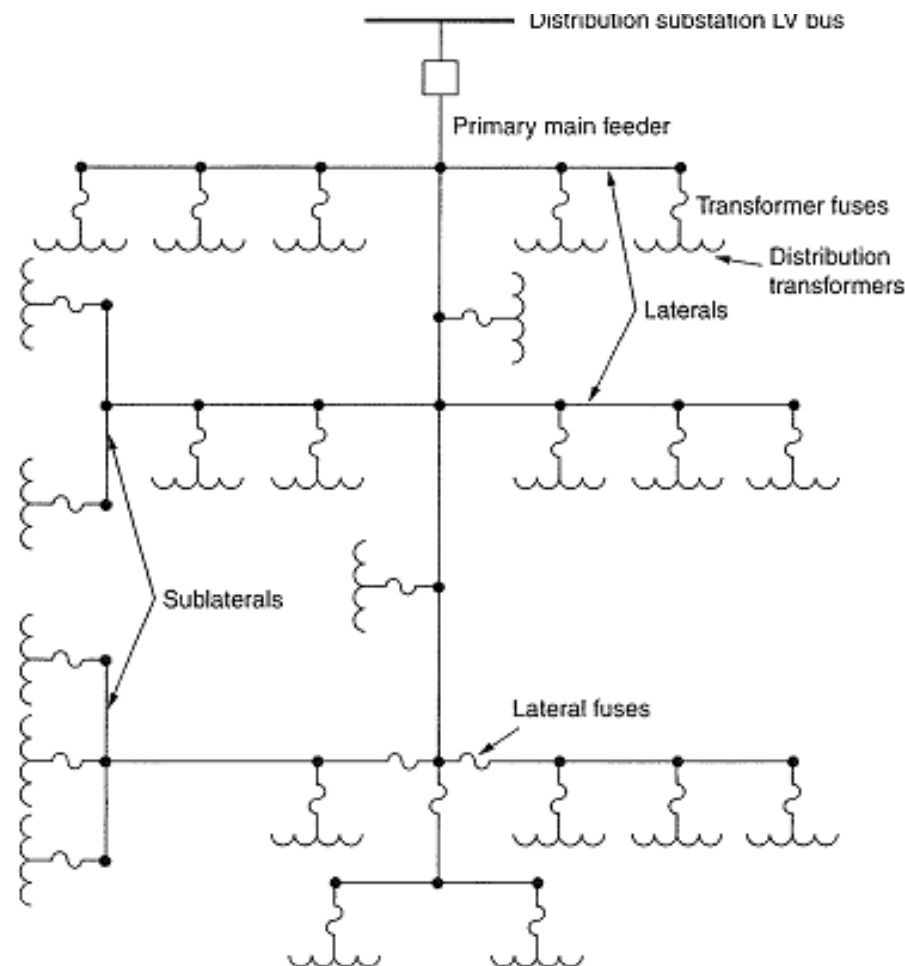


FIGURE 5.2 Radial-type primary feeder.

A primary fault causes the feeder breaker to be open. The breaker will remain open until the fault is isolated from both directions. The loop-type primary feeder arrangement is especially beneficial to provide service for loads where high service reliability is important. In general, a separate feeder breaker on each end of the loop is preferred, despite the cost involved. The parallel feeder paths can also be connected to separate bus sections in the substation and supplied from separate transformers. In addition to main feeder loops, normally open lateral loops are also used, particularly in underground systems.

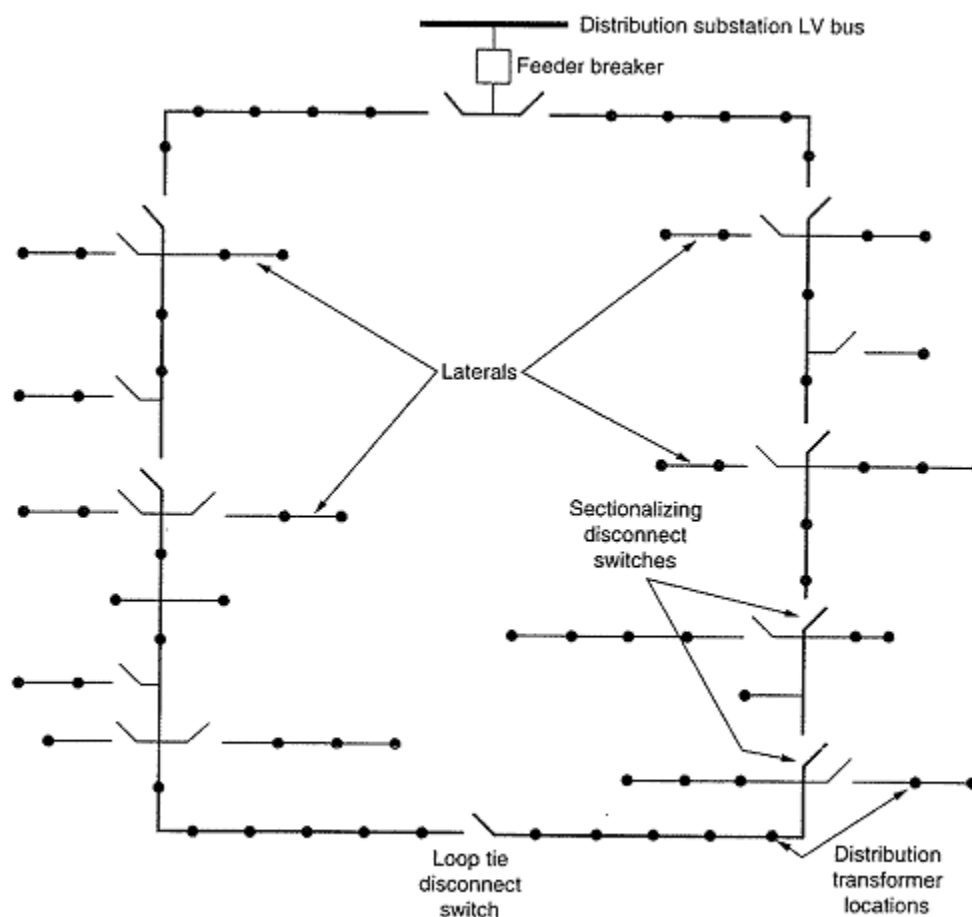
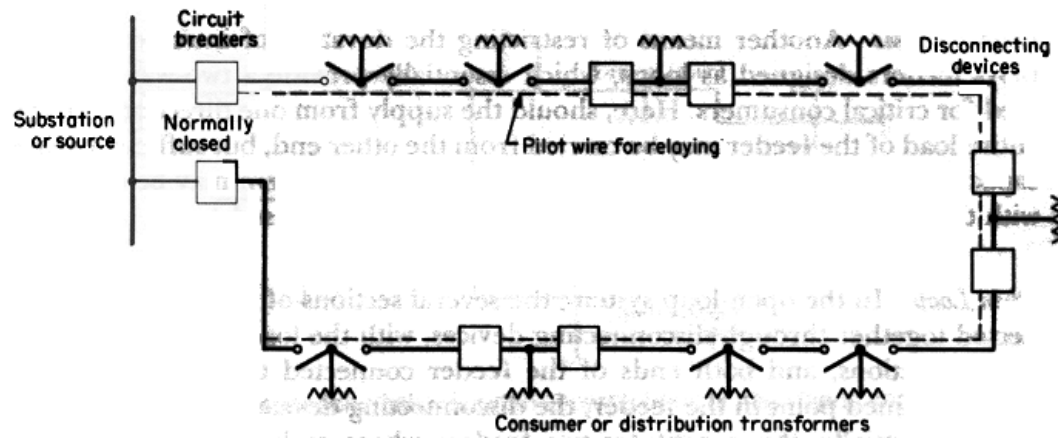
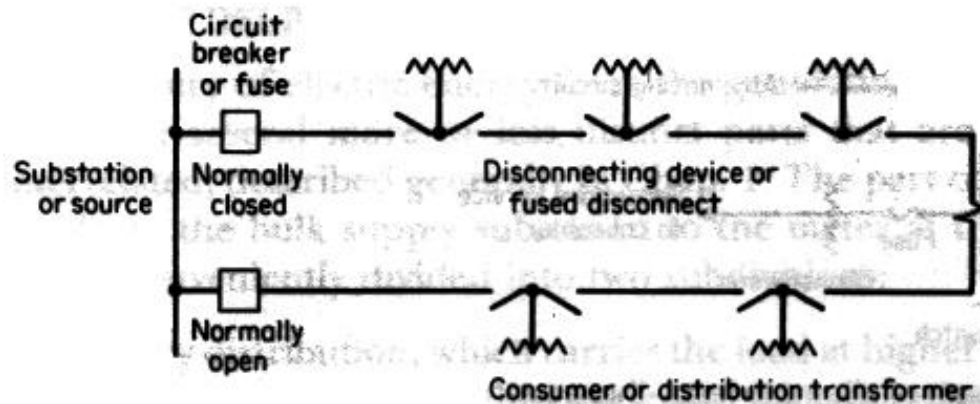


FIGURE 5.6 Loop-type primary feeder.



Closed-loop system schematic diagram



Open-loop system schematic diagram